

BRAC UNIVERSITY
Department of Computer Science and Engineering

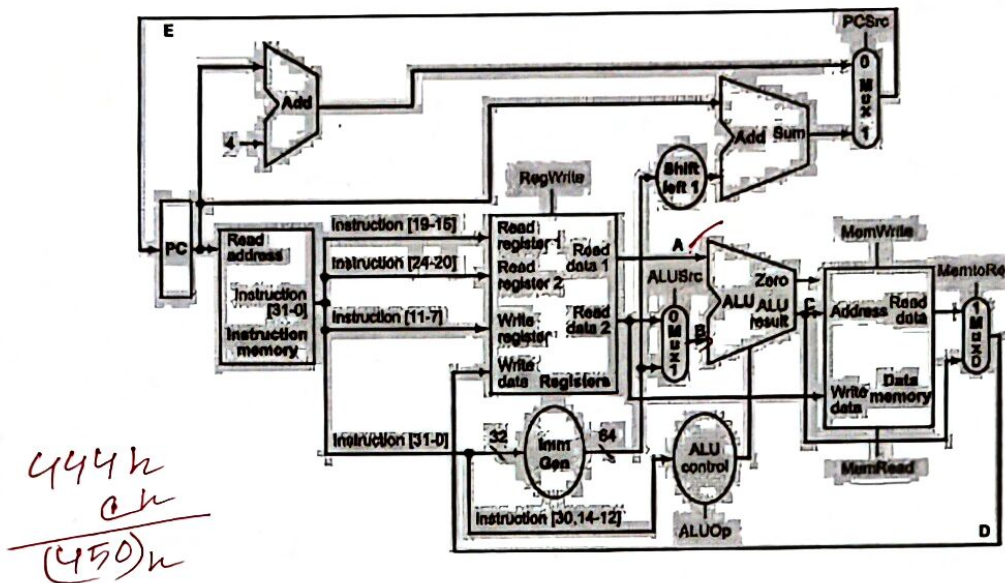
Examination: Quiz - 3
 Duration: 25minutes

Semester: Summer 2025
 Full Marks: 15

CSE 340: Computer Architecture

Name: <u>Solution</u>	ID:	Section:
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1.



Study the above RISC-V Datapath thoroughly. Assume that initially PC=600h, x1 = 222h, x2 = 444h, x3 = 666h, x4=999h. Determine the values of A,B,C,D,E in hex for the instructions executed in following order. If the value does not matter or can't be computed, write it as X (don't care)

$(12)_{10} = (1100)_2 = (C)_{16}$

	A	B	C	D	E
Ld x4, 12(x2)	(444)h	(12) ₁₀ / (C) ₁₆	(450)h	X	(604)h

2. True/False. In case a statement is False, provide the correct statement.

- In a pipelined RISC-V processor, the write-back stage is always required, for all the instructions. [1]
- In a single-cycle RISC-V processor, the clock cycle time is fixed by the slowest instruction. [1]
- A structural hazard can occur in a pipelined processor if the instruction memory and data memory shares the same hardware. [1]
- In the RISC-V datapath, the immediate generation unit is required only for load/store instructions. [1]
- In a pipelined processor, pipeline registers are placed between stages to prevent hazards. [1]

Answer:

a) True.

b) True.

c) True.

d) False. → Also required for Addi/Branch instructions.

e) False. → Pipeline registers exist to hold data between different pipeline stages.

3.

add X1, X2, X7

sub X2, X1, X5

ld X2, 16(X5)

sd X2, 16(X1)

sll X5, X6, X7

ld X2, 16(X5)

add X2, X1, X7

The above mentioned instruction is running on a pipelined datapath. Answer the following questions based on this.

3.

- ✓ add X1, X2, X7
- ✓ sub X2, X1, X5
- ✓ ld X2, 16(X5)
- ✓ sd X2, 16(X1)
- ✓ sll X5, X6, X7
- ✓ ld X2, 16(X5)
- add X2, X1, X7

The above mentioned instruction is running on a pipelined datapath. Answer the following questions based on this.

a) Solve the data hazard using only stalling. [4]

	cc1	cc2	cc3	cc4	cc5	cc6	cc7	cc8	cc9	cc10	cc11	cc12	cc13	cc14	cc15	cc16	cc17	cc18
(i)	IF	ID	EX	MEM	WB													
		X	X	X	X	X												
			X	X	X	X	X											
(ii)				IF	ID	EX	MEM	WB										
(iii)					IF	ID	EX	MEM	WB									
					X	X	X	X	X	X								
						X	X	X	X	X	X							
(iv)							IF	ID	EX	MEM	WB							
(v)								IF	ID	EX	MEM	WB						
(vi)									X	X	X	X	X	X				
(vii)										X	X	X	X	X	X			
(viii)											IF	ID	EX	MEM	WB			
												IF	ID	EX	MEM	WB		

b) How many clock cycles are needed to run the instruction set? [1]

Answer: 17

Instruction Fetch takes 15ps, Register read/write takes 15ps, ALU operation takes 30 ps, Memory access takes 15ps

c) Find the clock period. [1] Answer: 30ps

d) Total time to run the instruction. [1] Answer: $(17 \times 30) = 510 \text{ ps}$